Week 01_Module 01_Part 05

Saturday, November 8, 2025 2:01 PM

Distribution Shapes for ML: Symmetric vs Skewed, Long Tails, and IQR Outliers

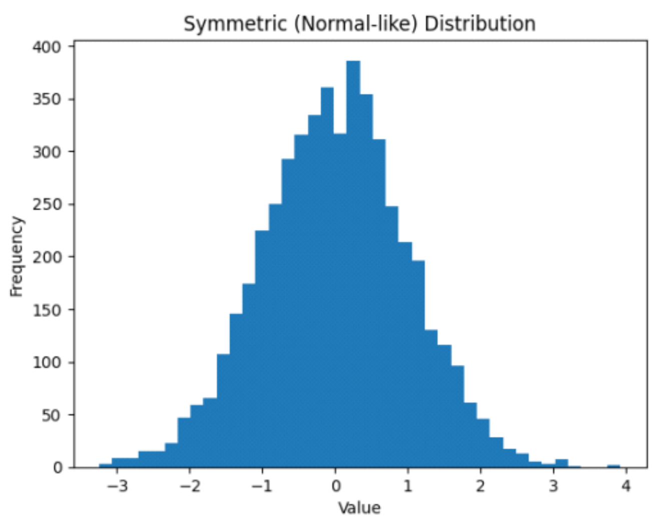
Goal: Understand how distribution shape affects ML models and practice outlier detection using IQR fences.

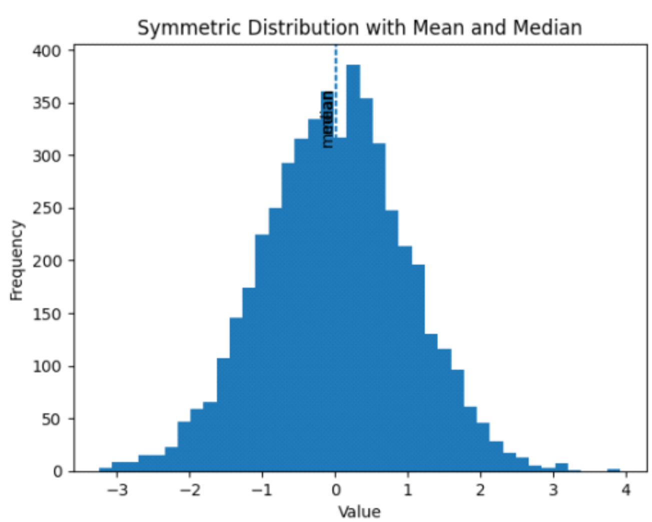
What you'll do:
- Visualize symmetric vs. skewed distributions
- See what long tails look like and why they matter
- Detect outliers using IQR fences in code

Symmetric Distributions

"A symmetric distribution looks the same on both sides of its center. The most common example is the **Normal Distribution**. Here, the mean, median, and mode all lie at the center."

Visuals:





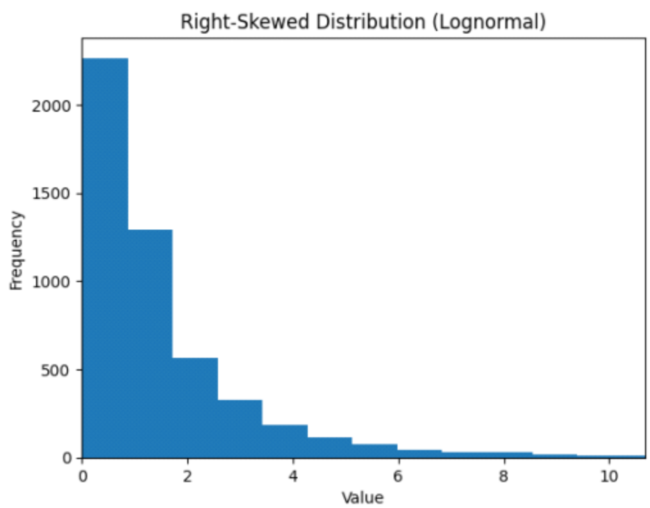
ML Focus:

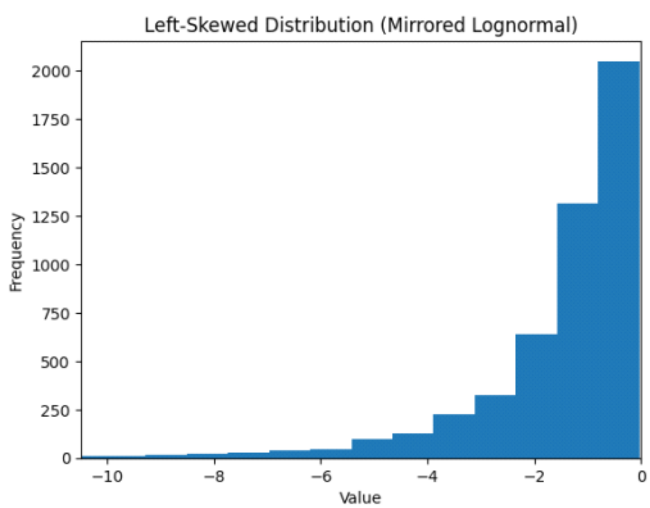
Many ML algorithms, such as **Linear Regression**, **SVMs**, and **KNN**, assume features follow a roughly normal distribution for optimal performance. If data is symmetric, we often don't need transformations.

Skewed Distributions

A skewed distribution is not symmetric. It can be **positively skewed (right-skewed)** or **negatively skewed (left-skewed)**.

Visuals:





ML Focus:

Right-skewed data often appears in income, price, or reaction time datasets. For ML models, skewness can reduce model accuracy and cause bias. To fix this, we often apply **log transformation**, **Box–Cox**, or **Yeo–Johnson transformations** to make data more symmetric.

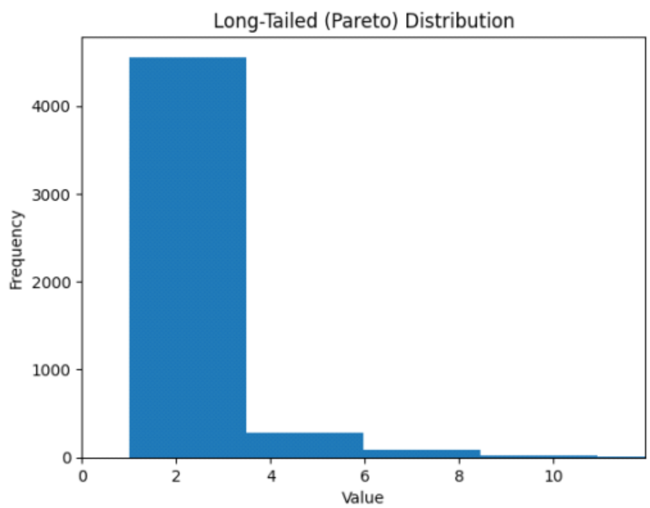
Example:

For instance, in predicting house prices, applying a log transform to the price column can improve model stability and performance.

Long Tails and Their Significance

A long tail means extreme values stretch far from the center. Long-tailed distributions are common in real-world datasets — like user engagement or social media followers — where a few points dominate the range.

Visuals:



ML Focus:

Long-tailed data challenges ML models. Models may overfit to frequent cases and ignore rare but important events — like fraud detection or rare disease prediction. Handling long tails often involves **data normalization**, **resampling**, or **outlier-aware models**.

In short:

Concept	Focus	Meaning	Example
Skewness	Asymmetry	Which side the data lean toward	Income distribution (right-skewed)
Long Tail	Tail length and thickness	How extended rare or extreme values are	YouTube video views (long right tail)

Outlier Detection using IQR Fences

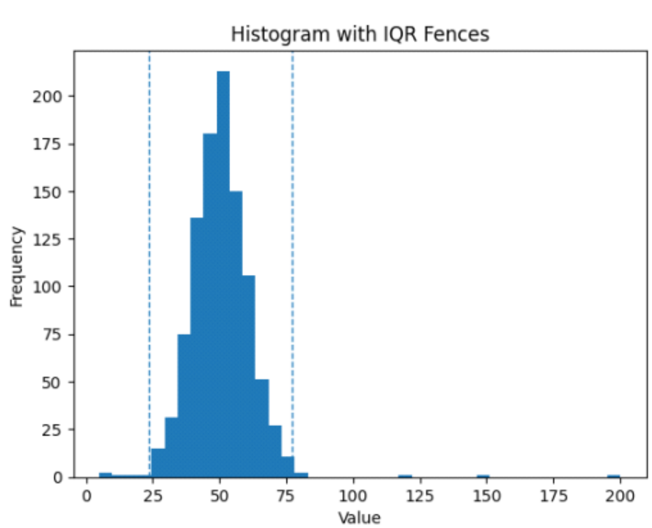
Outliers are data points that differ significantly from most observations. They can distort model training and degrade perfomance. A simple yet effective method to detect outliers is using the **Interquartile Range (IQR)**.

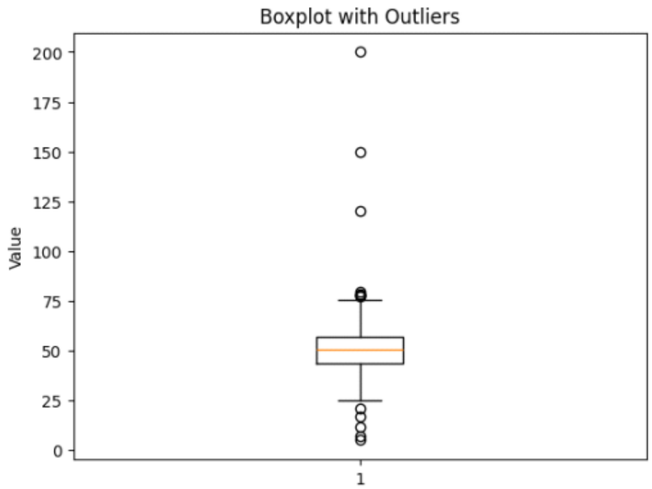
Step-by-step explanation:

1. Q1 = 25th percentile
2. Q3 = 75th percentile
3. IQR = Q3 – Q1
4. Lower Fence = Q1 – 1.5 × IQR
5. Upper Fence = Q3 + 1.5 × IQR

   Any data point outside these fences is considered an outlier.

Visuals:





- Q1 = 30, Q3 = 70 → IQR = 40
- Lower = 30 – 1.5×40 = –30, Upper = 70 + 1.5×40 = 130 → values < –30 or > 130 = outliers.

ML Focus:

In ML preprocessing, outlier removal or capping helps improve model robustness, especially for models sensitive to scale — like Linear Regression or KNN.

When to Keep vs. Remove Outliers

Not all outliers are bad. In some ML problems, like fraud detection or medical diagnosis, outliers represent critical cases. Instead of removing them, we may label or model them separately.

Recap & Summary

Let's summarize what we learned:

☑ Symmetric distributions make ML models more stable.
☑ Skewed or long-tailed data often need transformation.
☑ IQR is a quick and effective outlier detection method.
☑ Always analyze whether outliers represent noise or valuable insight.

Understanding data shape is the first step toward building trustworthy ML models.